

TECHNICAL BULLETIN**DIXIE CHEMICAL**

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**EPOXY CURING AGENT 100D
(ECA 100D)****Typical Properties**

Properties	Value	Test Method
Molecular Weight (average)	162 - 166	
Specific gravity @ 25°C, g/cc	1.15 – 1.21	D-102 (ASTM D 4052)
Appearance	Clear to light yellow liquid	D-104 (Visual)
Viscosity @ 25 °C, cP	50 - 100	D-150F
Color, Gardner	1 max	D-150G
Free acid, %	0.5 max	D-160B-3

Applications

ECA 100D is a clear to light yellow, low viscosity liquid epoxy curing agent designed for use in converting epoxy resins to highly crosslinked polymers with excellent physical and electrical properties. ECA 100 performs well when used in demanding applications, such as filament winding, pultrusion, and other forms of lamination, as well as electrical encapsulating, insulating, casting, and potting.

ECA 100D imparts ease of handling due to low viscosity, low toxicity and low volatility. It is a mixture, which is a liquid at normal room temperature. If the product crystallizes, it should return to liquid by heating 38-55°C (100-130°F) for 2-8 hours.

For more information on the use of anhydrides like ECA 100D as epoxy curing agents, please consult the Technical Bulletin, "Formulating Anhydride-Cured Epoxy Systems", available from Dixie Chemical.

The data contained herein are furnished for information only and are believed to be reliable. This information is provided only as guidance and is not to be considered a warranty or quality specification. Dixie Chemical makes no warranties (expressed or implied) as to its accuracy and assumes no liability in connection with any use of this product.

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ECA 100

Formulation Example

ECA 100D was formulated with a variety of epoxy resins. The samples were formulated with 100 parts of epoxy resin, 84-117 parts ECA 100, and 1 part 1-Methylimidazole. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

Epoxy type	ECA 100D, phr	Viscosity @ 25 °C, cP	Gel Time ¹ , min	Tg ² , °C
Standard BPA Liquid Epoxy	85	950	37	142
Low Viscosity BPA Liquid	88	690	40	143
Cycloaliphatic Epoxy	117	120	40	225
Epoxy Phenol Novolac	93	930	36	144
Epoxy BPA Novolac	84	3310	33	152
BPF Liquid Epoxy	95	490	38	130

1. Shyodu Hot Pot (100g @ 100°C)

2. Cured 1 hr @ 120°C and 1 hr @ 220°C. Analyzed in DSC: 40°C/min to 250°C

Handling & Storage

ECA 100D will react with water to form diacids. This is normally undesirable, so ECA 100D should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for ECA 100D, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

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Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes.

In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 100D is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie representative for details.